

Paper	ENCN342 – Fluid Mechanics and Hydraulics
Topic	Steady Open Channel Flow
Module	4 – Uniform Flow

Summary

This module presents the concept of uniform flow.

Prerequisites

Module 3

Objectives

At the conclusion of this module you should:

1. Be able to define uniform flow.
2. Understand how the momentum equation can have friction and gravity included in it, and be able to write down the momentum equation for this case.
3. Be familiar with the Chezy and Manning resistance equations.

References

Hunt 1995 pp 12.12-12.32

Key Points

1. Uniform flow

Uniform flow is a flow state in which the flow depth and velocity do not change with distance downstream. It occurs when the driving gravitational force along the channel exactly balances the frictional forces on the channel boundary.

2. Resistance laws

When friction and gravity are included in the momentum equation the momentum equation can be written as

$$\tau = -\rho g R \left(\frac{1}{2g} \frac{du^2}{dx} + \frac{dy}{dx} + \frac{dz}{dx} \right) = -\rho g R \frac{dH}{dx} = \rho g R S_f \quad (1)$$

where τ is the average shear stress on the boundary and R is the **hydraulic radius** defined to be

$$R = \frac{A}{P} \quad (2)$$

where P is the **wetted perimeter** of the channel, and A is its cross-sectional area. S_f is termed the **friction** or **energy slope**.

When uniform flow occurs equation (1) becomes

$$\tau = -\rho g R \frac{dz}{dx} = \rho g R S_0 \quad (3)$$

where S_0 is the bed slope.

A number of resistance laws can be used to relate the shear stress to the flow variables, y and u .

2.1 Chezy Equation

Chezy's equation is

$$u = C \sqrt{RS} \quad (4)$$

where C is a constant that can be related to the Darcy friction factor by

$$C = \sqrt{\frac{8g}{f}} \quad (5)$$

2.2 Manning Equation

Manning's equation is

$$u = \frac{R^{2/3} S^{1/2}}{n} \quad (6)$$

where n is a parameter that increases with boundary roughness. It is related to the Chezy C by

$$C = \frac{R^{1/6}}{n} \quad (7)$$

3. Uniform flow as a control

Uniform flow, because it imposes a relationship between the flow depth and the flow velocity through one of the resistance laws, can be considered a flow control, even though it is not a localised feature in the channel. However, when other controls are present in the channel, as they invariably are, the flow is pulled away from uniform conditions.

Worked Examples

Question 1

Find the slope required to pass 10 cumecs through a rectangular channel that is 3m wide and lined with unfinished concrete, so that the Froude number is not less than 2.

Solution

We will use Manning's law to determine the slope for this channel. In this case we know the velocity and hydraulic radius, and we are wanting to find the slope S_0 . So rearranging equation 6 we have

$$S = \left(\frac{un}{R^{2/3}} \right)^2 = \left(\frac{Qn(B+2y)^{2/3}}{yB(yB)^{2/3}} \right)^2 = \left(\frac{Qn(B+2y)^{2/3}}{(yB)^{5/3}} \right)^2$$

The channel is to be lined with unfinished concrete. Referring to Henderson (or any other hydraulics book with a table of Manning n values) we find n is roughly 0.014.

At this point S cannot be determined unless the flow depth is known. The second piece of information in the question is that the flow must be supercritical and cannot have a Froude number less than 2. Therefore the minimum slope will correspond to a Froude number of 2. If Fr is greater than this the flow depth will be less and clearly the channel slope would need to be greater according to the above equation.

For the discharge and width given the flow depth for $Fr = 2$ is

$$y = \sqrt[3]{\frac{q^2}{gFr^2}} = \sqrt[3]{\frac{(Q/B)^2}{4g}} = \sqrt[3]{\frac{(10/3)^2}{4 \times 9.81}} = 0.65m$$

Now we can calculate the slope

$$S_0 = \left(\frac{Qn(B+2y)^{2/3}}{(yB)^{5/3}} \right)^2 = \left(\frac{0.014 \times 10 \times (3 + 2 \times 0.65)^{2/3}}{(0.65 \times 3)^{5/3}} \right)^2 = 0.015$$

This is a very steep slope. Note the velocities in the channel are roughly 5.2m/s.

Question 2

A canal of rectangular section and $n = 0.014$ is to be laid on a slope of 0.001 and is to carry a discharge of 100 cumecs. Calculate the required waterway area for the following values of the ratio B/y : 1, 1.5, 2, 2.5, 3, 3.5.

Solution

The waterway area can be found from Manning's friction law and some algebraic manipulation. Manning's law states

$$u = \frac{R^{2/3} S^{1/2}}{n}$$

or

$$\frac{Q}{By} = \frac{\left(\frac{By}{B+2y} \right)^{2/3} S^{1/2}}{n}$$

Thus

$$By \left(\frac{By}{B+2y} \right)^{2/3} = \frac{nQ}{S^{1/2}}$$

Dividing both top and bottom lines inside the bracket by y , and setting the ratio B/y to α , we get

$$By \left(\frac{B}{2+\alpha} \right)^{2/3} = \frac{nQ}{S^{1/2}}$$

or

$$B^{5/3} y = \frac{nQ(2+\alpha)^{2/3}}{S^{1/2}}$$

or

$$B^{8/3} = \frac{\alpha nQ(2+\alpha)^{2/3}}{S^{1/2}}$$

All quantities on the right hand side of this equation are known, and therefore B can be calculated. Once B is known the flow depth can be calculated by the choice of α see table 1.

Table 1: Tabulated values of α , B , y and A .

α	B (m)	y (m)	$A=By$ (m ²)
1	5.45	5.45	29.7
1.5	6.60	4.40	29.0
2	7.60	3.80	28.9
2.5	8.51	3.40	29.0
3	9.35	3.12	29.2
3.5	10.15	2.90	29.4

The flow area changes very little in these calculations. Thus as the channel widens the flow depth adjusts so that the flow area, and flow velocity are roughly unchanged.